

Заг. 1. $X_1 = \# \text{співхоро в отдор } A^n$
 $\sim \text{Bm}(20, \frac{1}{2}) \Rightarrow \boxed{EX_1 = 10}$.

$X_2 = \# \text{гравки в отдор}$
 $\sim \text{Bm}(10, \frac{1}{2}) \Rightarrow \boxed{EX_2 = 5}$

2. $P(\text{конкретна гра в отдор є партнером})$
 $= \frac{1}{2} \Rightarrow \text{отг.} = \boxed{EGe(\frac{1}{2}) = 2}$

3. $P(\text{бавки гравки гра са загро}) = \left(\frac{1}{2}\right)^{10}$
 $\Rightarrow \text{отг.} = \boxed{EGe\left(\left(\frac{1}{2}\right)^{10}\right) = 1024}$.

Ако X_A и X_B са $\# \text{гравки в}$
отдор A и B , $X_A \neq X_B$ и
 $X_A + X_B = 10 \Rightarrow \boxed{EX_A = 5}$.

Заг. 2 $H_i = \sum \text{терми } i \text{ стандартно } \cup$
2-й тесту.

$A = \{2 \text{ б-ти от } 2 \text{ взвращения}\}$

A priori: $P(H_2) = \frac{1}{4}$; $P(H_1) = \frac{1}{2}$; $P(H_0) = \frac{1}{4}$

$P(A|H_2) = \left(\frac{1}{6}\right)^2$; $P(A|H_1) = 40\% \cdot \frac{1}{6}$; $P(A|H_0) = \left(\frac{40\%}{6}\right)^2$

$$P(H_2|A) = \frac{\left(\frac{1}{6}\right)^2 \cdot \frac{1}{4}}{\left(\frac{1}{6}\right)^2 \cdot \frac{1}{4} + \frac{1}{2} \cdot 40\% \cdot \frac{1}{6} + \frac{1}{4} \cdot \left(\frac{40\%}{6}\right)^2} = \frac{25}{289} = \left(\frac{5}{17}\right)^2$$

$$P(H_1|A) =$$

$$P(H_0|A) = \dots = \left(\frac{12}{17}\right)^2$$

т.е. $P(\text{не стандартно } \text{заг} | A) = \frac{12}{17}$
 $\approx 70.59\%$

Можем да подходим и по 2 етапа:

$A = \{ \text{теглящ зор и хвърляне 6} \}$

$H_0 = \{ \text{теглящ стандартен} \} = H_1$

$$P(H_0) = P(H_1) = 1/2$$

$$P(A|H_0) = \frac{1}{6} \quad ; \quad P(A|H_1) = 40\%$$

$$\Rightarrow \underbrace{P(H_0|A)}_{H_0^*} = \frac{1/6 \cdot 1/2}{1/6 \cdot 1/2 + 2/5 \cdot 1/2} \quad ; \quad \underbrace{P(H_1|A)}_{H_1^*} = \frac{12}{17}$$

$$P(H_0^*) = \frac{5}{17} \approx 29,41\%$$

$$\text{и следователно } P(H_0^*|A) = \frac{1/6 \cdot 5/17}{1/6 \cdot 5/17 + 2/5 \cdot 12/17} = \frac{25}{169} \\ \approx 14,79\% \quad \text{и съответно } \approx 85,21\% \quad \text{за втория етап}$$

* Ако изтегляме n пъти и хвърляме всеки път 6 и

$H_i = \{ \text{изтегляне на } i \text{ стандартни} \}$

$$P(H_i) = \binom{n}{i} \left(\frac{1}{2}\right)^n$$

$$P(A|H_i) = \left(\frac{1}{6}\right)^i \left(\frac{2}{5}\right)^{n-i}$$

$$\Rightarrow P(H_n|A) = \frac{\binom{n}{n} \left(\frac{1}{2}\right)^n \left(\frac{1}{6}\right)^0 \left(\frac{2}{5}\right)^{n-0}}{\sum_{i=0}^n \binom{n}{i} \left(\frac{1}{2}\right)^n \left(\frac{1}{6}\right)^i \left(\frac{2}{5}\right)^{n-i}} = \frac{\left(\frac{1}{6}\right)^n}{\left(\frac{17}{30}\right)^n} = \left(\frac{5}{17}\right)^n$$

$$P(\text{tangeren}) = \frac{5}{11} \quad ?!$$

Прогресс за
унапредување!

200.3 1. Трябва всеки път или да се
включи едновременно, или да се произведе
едновременно.
⇒ отг. е $(75\% \cdot 80\% + 25\% \cdot 20\%)^5$

= 11,6 %

2. Για οβε απορ 1 οτ 1-1ε 5 : $X \sim \mathcal{B}_n(5, 80\%)$
— 11— 2 — 11— : $Y \sim \mathcal{B}_n(5, 75\%)$

$$\begin{aligned} P(X=Y) &= \sum_{k=0}^5 P(X=Y=k) \\ &= \sum_{k=0}^5 \binom{5}{k} (25\% \cdot 80\%)^k (25\% \cdot 20\%)^{5-k} \\ &= 0.2984 \end{aligned}$$

$$P(\text{pa bekommt auch richtige ID}) \\ = P(X=4) \cdot (75\% \cdot 80\% + 25\% \cdot 20\%) \\ \approx 3,46\%$$

$$3. E_{\text{выигрыш}} = 10 \cdot (1 - P(X=Y)) + P(X=Y) \left(10 + \frac{1}{1 - (75\% \cdot 80\% + 25\% \cdot 20\%)} \right) \\ \approx 10,85.$$



ИЗДАНИЕ $X_1 \sim \mathcal{U}(0,1)$
и $Y \sim \mathcal{U}(0, 1-X_1)$

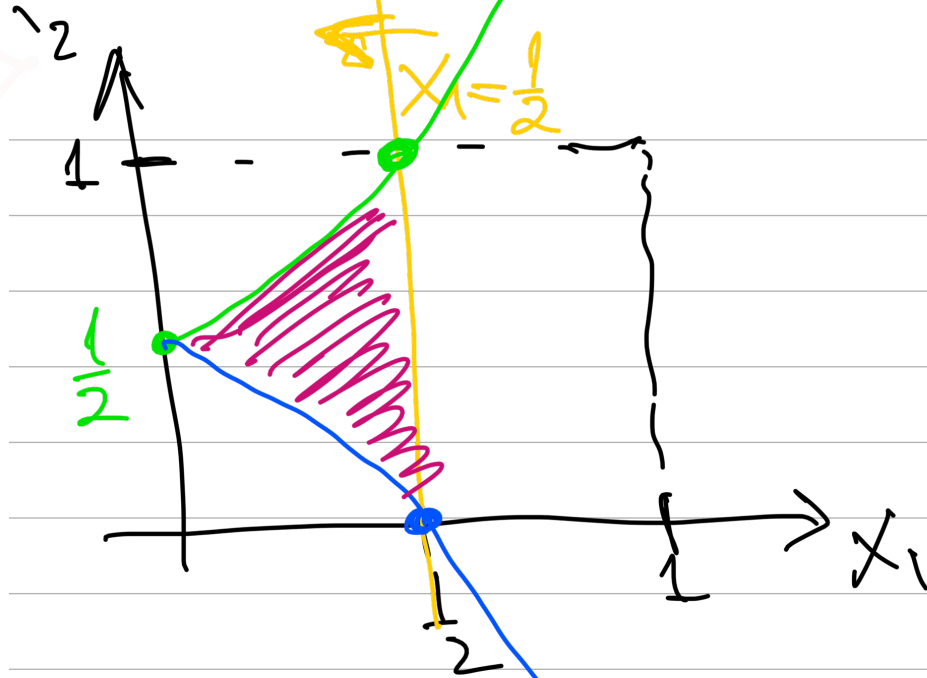
Можно да заменим Y като $(1-X_1)X_2$,
когато $X_2 \sim \mathcal{U}(0,1)$ и $X_1 \perp X_2$.

Условието $X_1 + Y > 1 - X_1 - Y$ $\Leftrightarrow$ $X_1 + Y > \frac{1}{2}$

$$\begin{cases} X_1 + 1 - X_1 - Y > Y \\ Y + 1 - X_1 - Y > X_1 \end{cases} \Leftrightarrow \begin{cases} Y < \frac{1}{2} \\ X_1 < \frac{1}{2} \end{cases}$$

$$\Leftrightarrow \begin{cases} X_1 + (1-X_1)X_2 > \frac{1}{2} \\ (1-X_1)X_2 < \frac{1}{2} \\ X_1 < \frac{1}{2} \end{cases} \Leftrightarrow \begin{cases} X_2 > \frac{1-2X_1}{2(1-X_1)} \\ X_2 < \frac{1}{2(1-X_1)} \\ X_1 < \frac{1}{2} \end{cases}$$

$$\Rightarrow X_2 = \frac{1}{2(1-X_1)}$$



$$x_2 = \frac{1-2x_1}{2(1-x_1)}$$

$$D_{\pi} = \frac{\int_0^{1/2} \left(\frac{1}{2(1-x)} - \frac{1-2x}{2(1-x)} \right) dx}{\int_0^{1/2} \frac{1}{2(1-x)} dx} = \frac{1}{\int_0^{1/2} \frac{x}{1-x} dx}$$

$$= \ln 2 - \frac{1}{2} \approx 19,32\%$$